

APPLICABILITY OF DECISION TREES FOR SITE SPECIFIC IRRIGATION MANAGEMENT

by

Trevor Askins

1

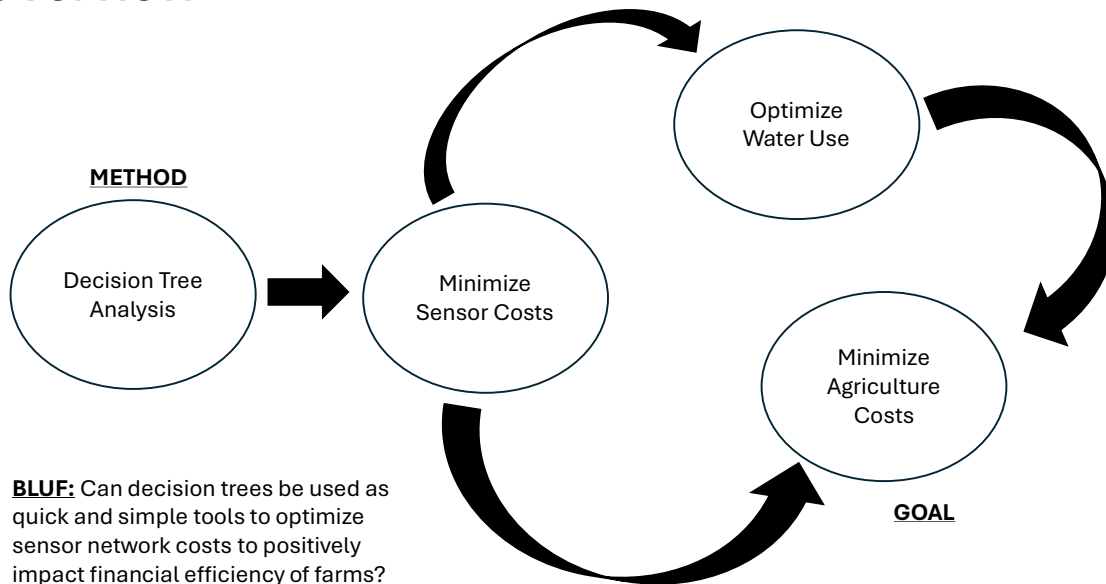
1

Trevor Askins



2

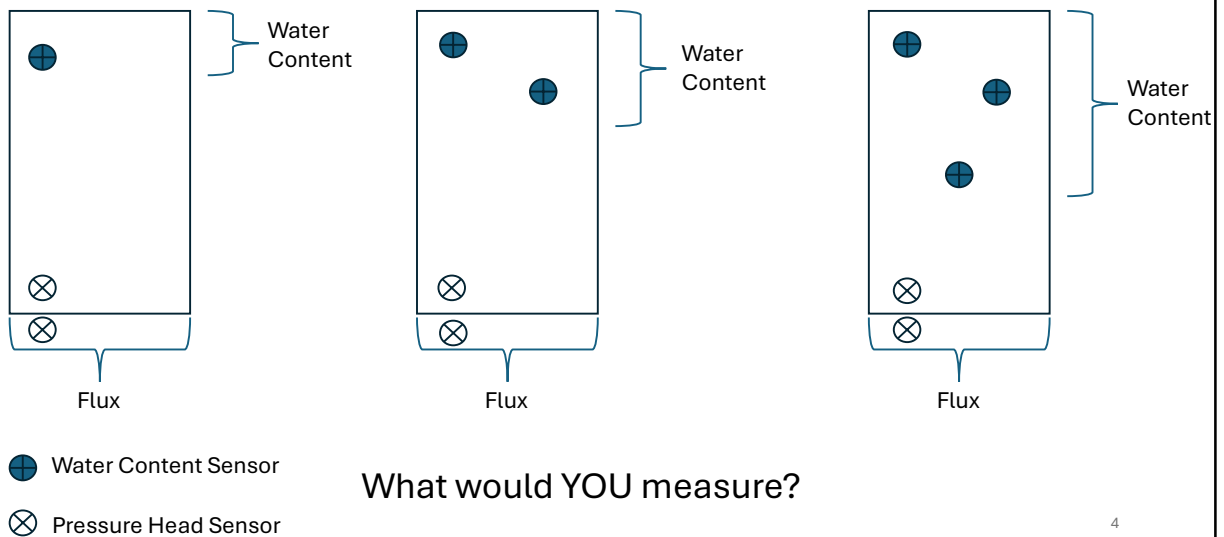
Overview



3

3

Sensor Options



4

4

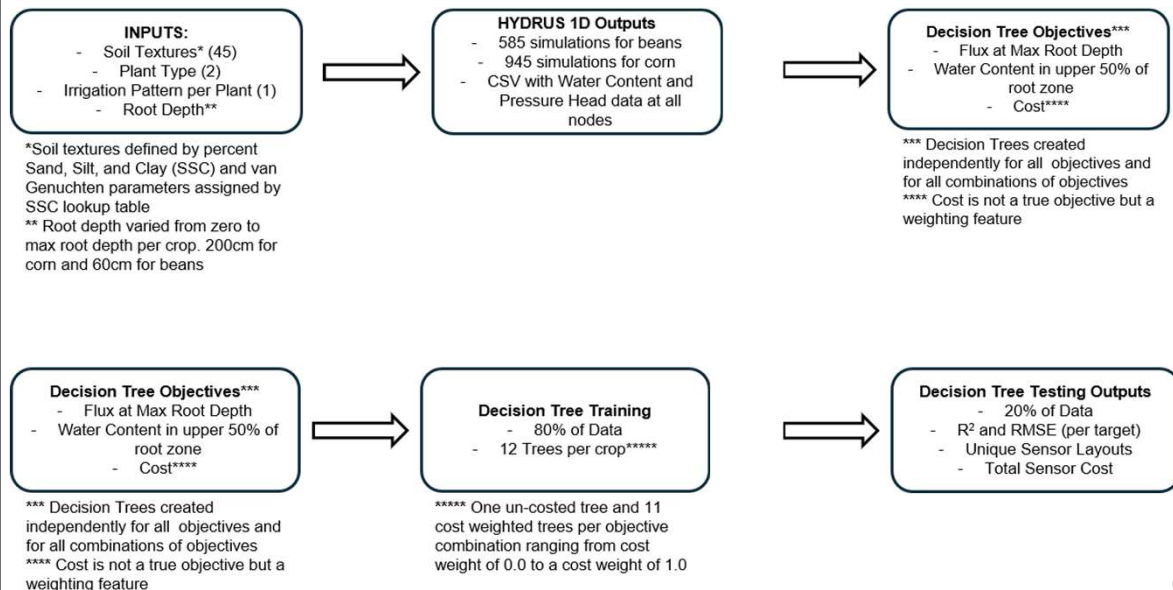
Why Decision Trees?

- What would YOU measure?
- 50 Depths
- 2 Sensor Types
- 1,300,000,000,000,000,000,000,000,000 Sensor Combinations
(One nonillion three hundred octillion)
- Per Objective (6)

5

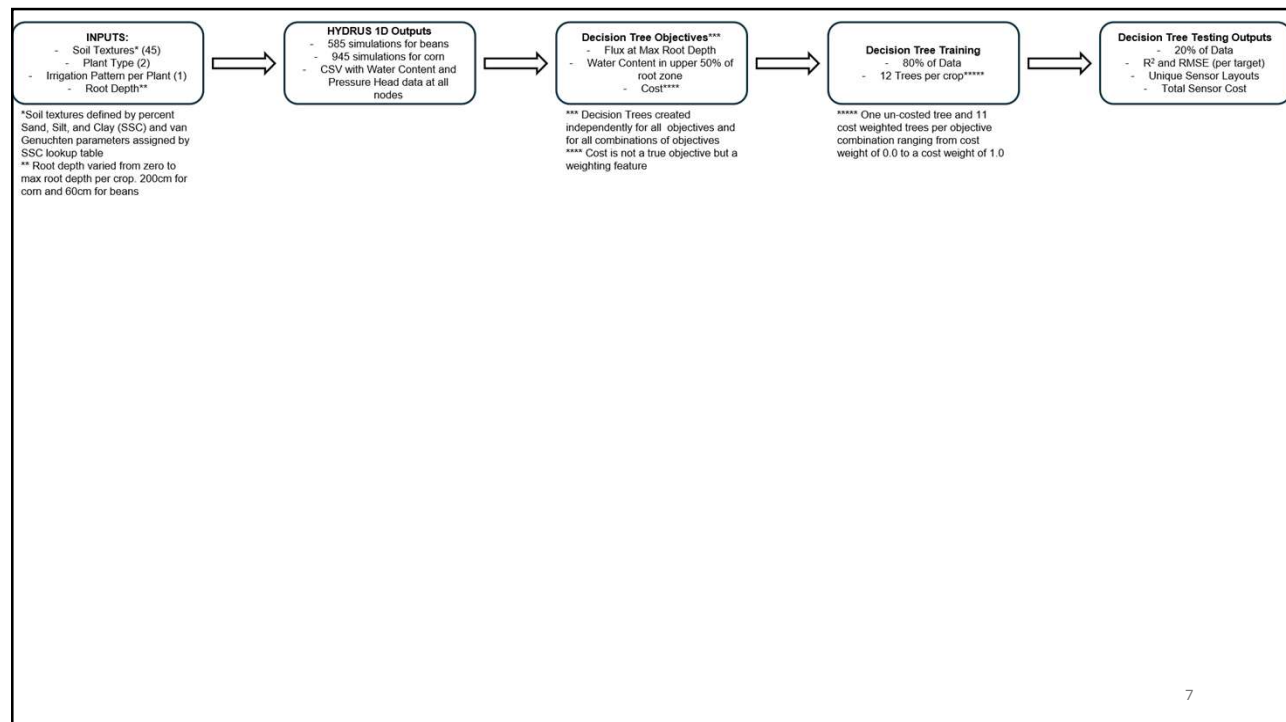
5

Method Summary

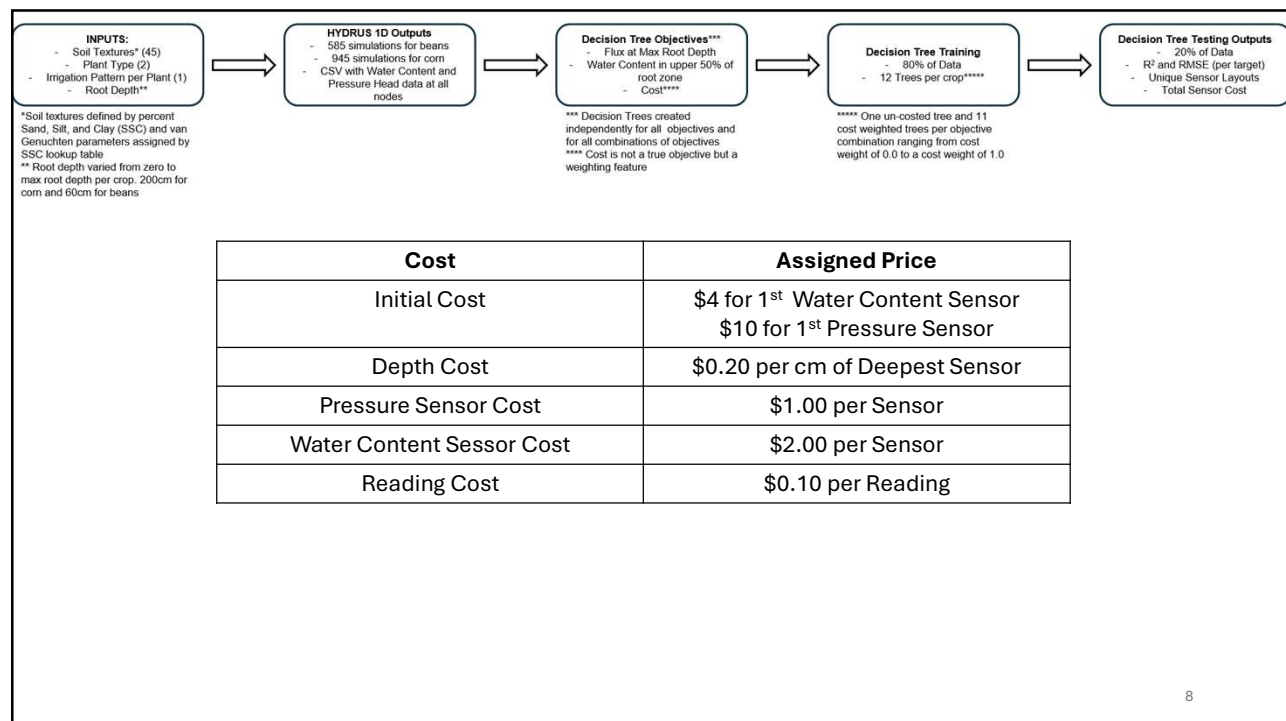


6

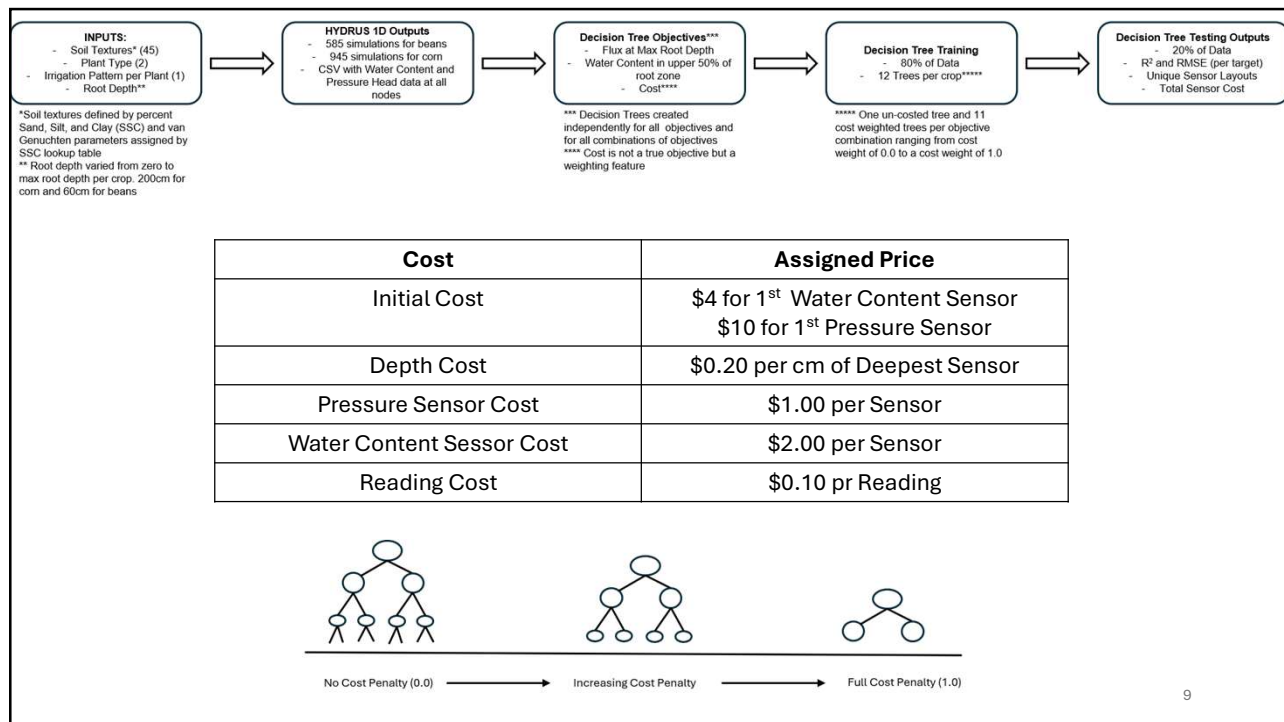
6



7

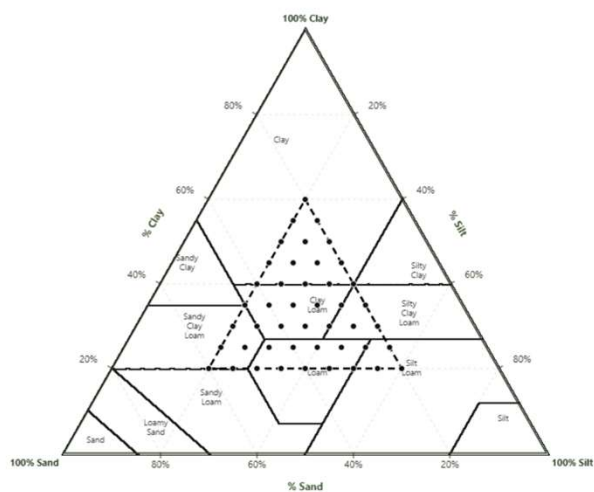


8



9

Hydrus



1 Experiment: Flood Irrigation

945 Simulations: 45 soils x 21 root depths

1600 Data Points per Simulation:
 2 Sensor Types x 50 Depths x 16 times

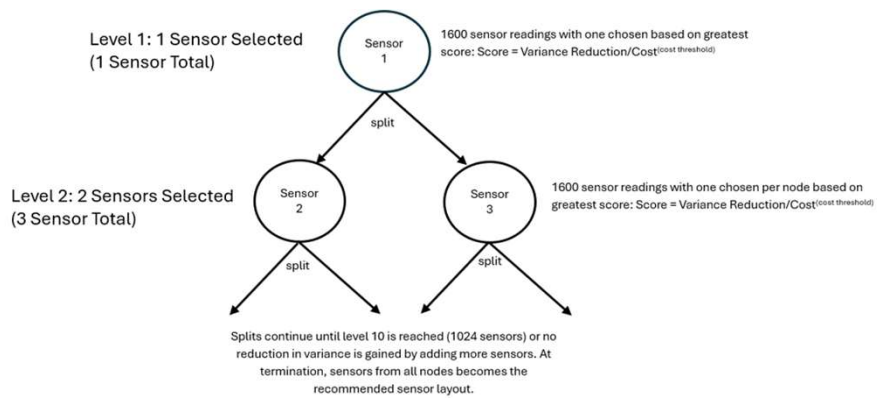
1 Data Point = 1 Sensor (Type, Time, Depth)

Crop	Corn	Beans
Initial Head Boundary (0-24 hours)	12cm	10cm
Secondary Head Boundary (24-48 hours)	6cm	5cm
Period of Zero Flux	15 Days	7 Days
Max Root Depth	200cm	60cm

10

10

Sorting Simulations



Sensor	Type	Depth	Variance Reduction	Marginal Cost
A	Pressure (h)	176cm	1000	\$10.00
B	Water Content (θ)	16cm	750	\$5.00

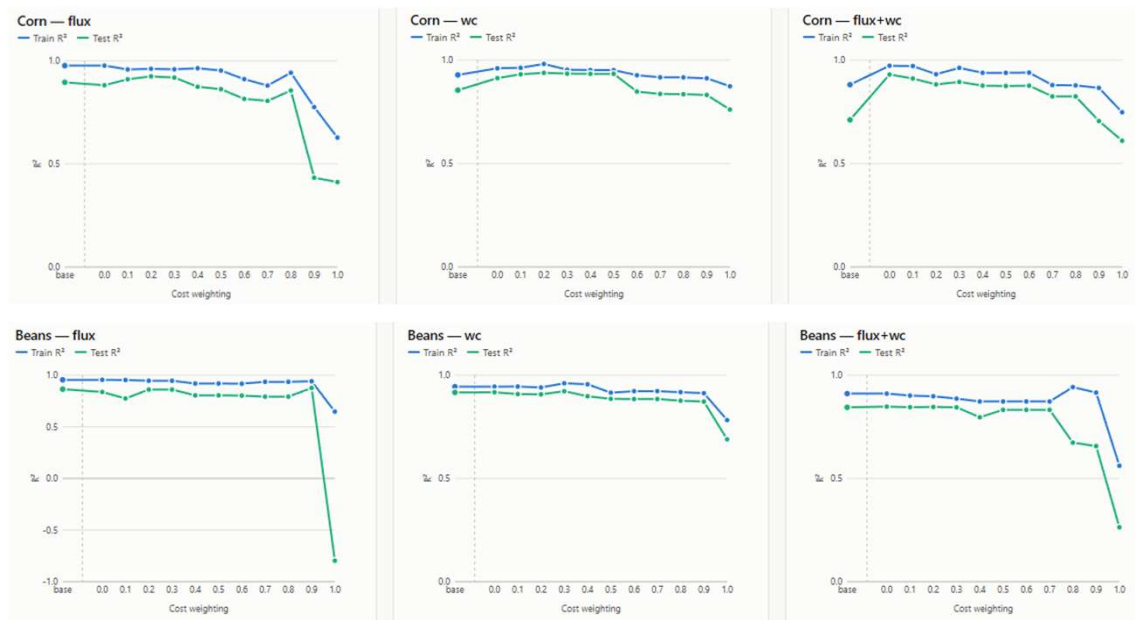
	Score A Equation	Score A Raw	Score B Equation	Score B Raw
Cost Weight: 0.0	$1000/10^{0.0}$	1000	$750/5^{0.0}$	750
Cost Weight: 0.5	$1000/10^{0.5}$	316	$750/5^{0.5}$	335
Cost Weight: 1.0	$1000/10^{1.0}$	100	$750/5^{1.0}$	150

Score = Variance Reduction

Score = Variance Reduction / $\text{Cost}^{(\text{cost threshold})}$

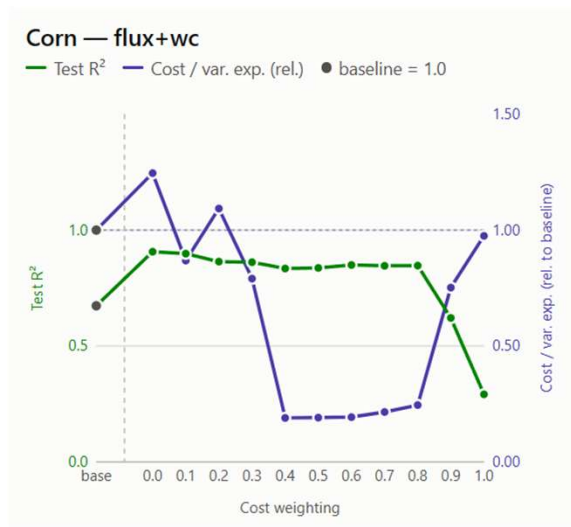
11

11



12

12



13

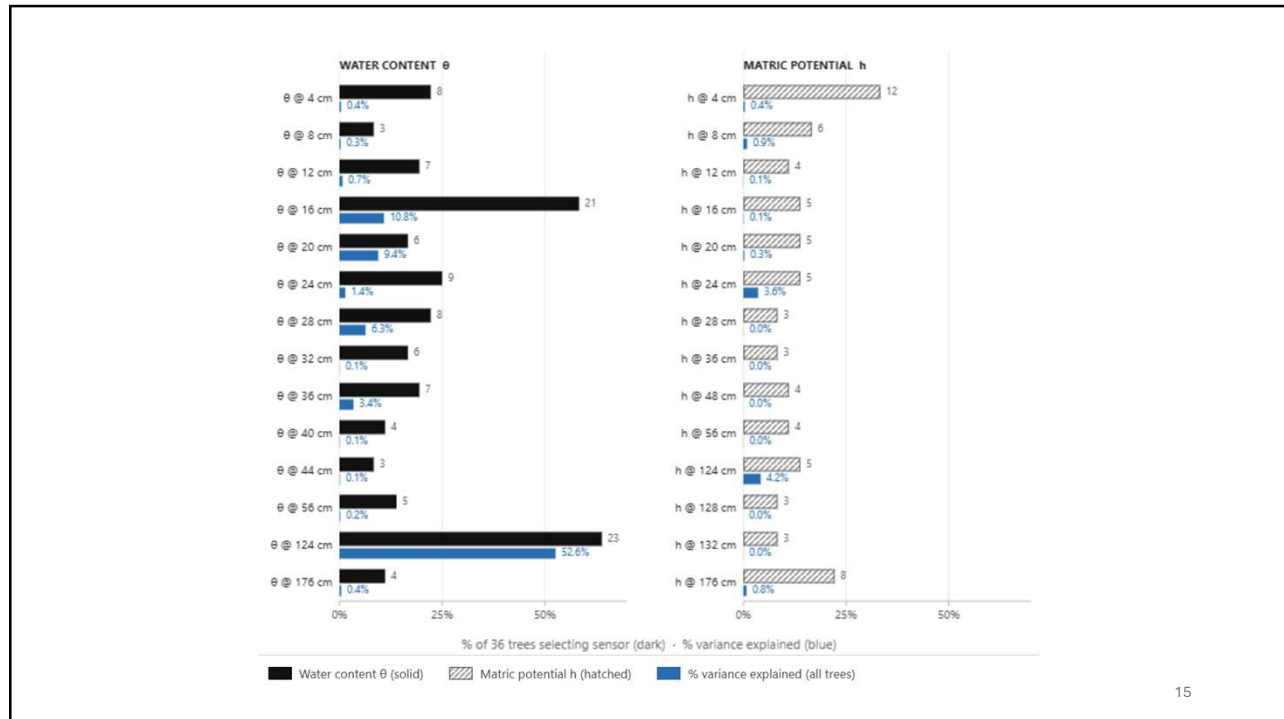
13

Corn Flux+WC Content								
Cost Threshold	Combined R2	Flux R2	WCR2	R2 Gap (flux-wc)	Flux RMSE	WC RMSE	Cost	Sensors
none	0.7109	0.5665	0.8553	-0.2888	0.002829	0.0102	65.6	19 (10wc@4-172cm, 9p@4-112cm)
0	0.93	0.8971	0.9629	-0.0658	0.001378	0.0051	91.4	28 (13wc@4-64cm, 15p@4-164cm)
0.1	0.9108	0.8526	0.9689	-0.1163	0.00165	0.0047	64.8	22 (9wc@4-124cm, 13p@4-140cm)
0.2	0.8822	0.8085	0.9559	-0.1474	0.00188	0.0056	79.5	17 (7wc@4-124cm, 10p@4-176cm)
0.3	0.8942	0.8365	0.952	-0.1155	0.001738	0.0059	56.8	10 (6wc@4-124cm, 4p@4-100cm)
0.4	0.8763	0.7942	0.9584	-0.1642	0.001949	0.0054	14.5	2 (2wc@16-124cm)
0.5	0.8748	0.7903	0.9594	-0.1691	0.001968	0.0054	14.6	2 (2wc@16-124cm)
0.6	0.8762	0.7941	0.9582	-0.1641	0.00195	0.0055	14.8	2 (2wc@16-124cm)
0.7	0.8242	0.8033	0.8452	-0.0419	0.001906	0.0105	14.3	2 (2wc@16-124cm)
0.8	0.8242	0.8027	0.8456	-0.0429	0.001909	0.0105	16.3	2 (2wc@16-124cm)
0.9	0.7053	0.568	0.8426	-0.2746	0.002825	0.0106	40.1	4 (3wc@36-124cm, 1p@56cm)
1	0.61	0.4323	0.7878	-0.3555	0.003238	0.0123	42	4 (3wc@36-124cm, 1p@56cm)

Observed Flux: -0.0175 – 0.00284 cm/cycle (16 days)
 Observed WC: 0.198 – 0.341

14

14



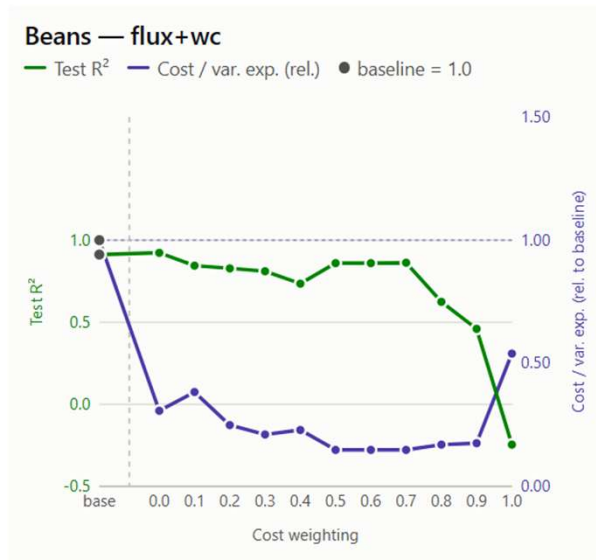
15

Corn Takeaway

- Cost Efficiency between 0.4 and 0.8
- Heavy Reliance on Deep Water Content Sensors

16

16



17

17

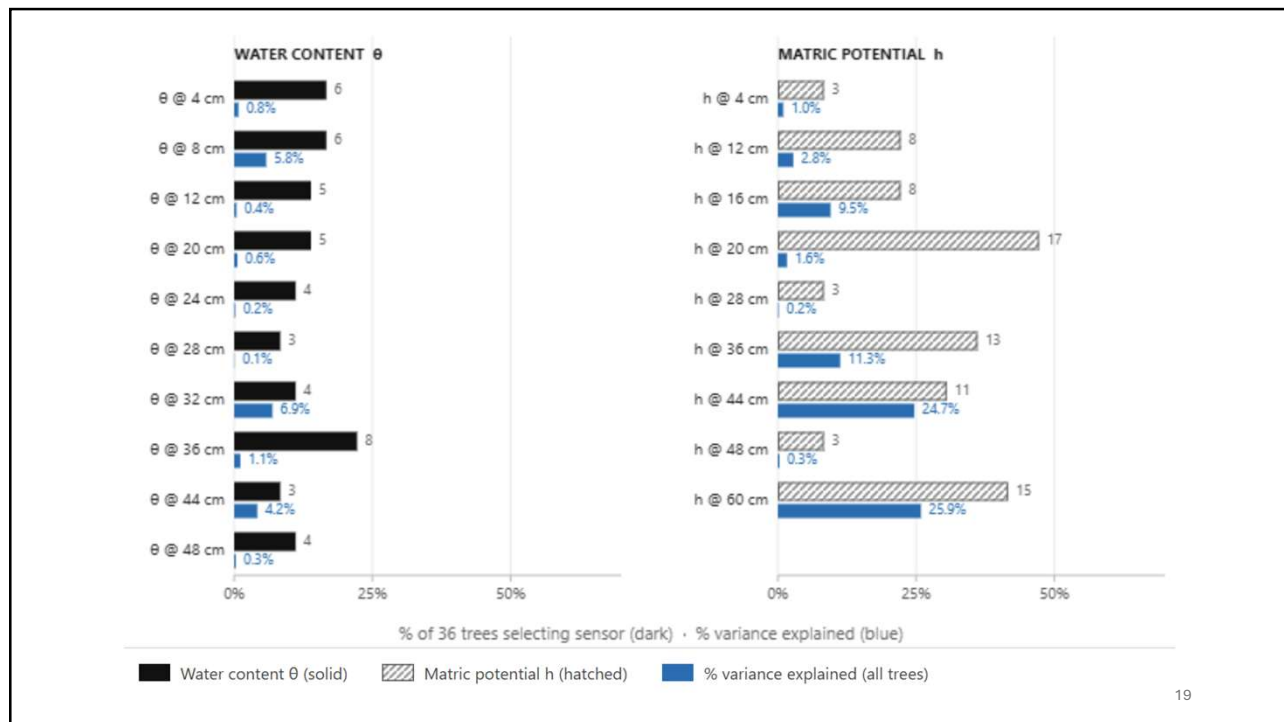
Beans Flux+WC								
Cost Threshold	Combined R2	Flux R2	WC R2	R2 Gap (flux-wc)	Flux RMSE	WC RMSE	Cost	Sensors
none	0.844	0.8491	0.8389	0.0102	0.00798	0.01197	91.5	13 (9wc@8-200cm, 4p@20-184cm)
0	0.8475	0.8578	0.8371	0.0207	0.00775	0.01204	28.1	6 (2wc@8-32cm, 4p@20-48cm)
0.1	0.8448	0.8309	0.8586	-0.0277	0.00845	0.01121	35	4 (3wc@8-44cm, 1p@36cm)
0.2	0.8459	0.8157	0.8761	-0.0604	0.00883	0.0105	22.7	3 (1wc@36cm, 2p@36-60cm)
0.3	0.8441	0.8365	0.8518	-0.0153	0.00831	0.01148	19.1	2 (1wc@36cm, 1p@36cm)
0.4	0.796	0.7658	0.8263	-0.0605	0.00995	0.01243	20.4	3 (1wc@12cm, 2p@12-44cm)
0.5	0.8319	0.8342	0.8297	0.0045	0.00837	0.01231	13.2	1 (1p@44cm)
0.6	0.8319	0.8342	0.8297	0.0045	0.00837	0.01231	13.2	1 (1p@44cm)
0.7	0.8319	0.8342	0.8297	0.0045	0.00837	0.01231	13.2	1 (1p@44cm)
0.8	0.6732	0.4596	0.8868	-0.4272	0.01511	0.01003	15.5	2 (2p@16-60cm)
0.9	0.6565	0.4195	0.8935	-0.474	0.01566	0.00973	15.1	2 (2p@16-60cm)
1	0.2637	-0.1663	0.6936	-0.8599	0.0222	0.0165	14.6	2 (2p@16-60cm)

Observed Flux: -0.219 – 0.057 cm/cycle (8 Days)

Observed WC: 0.258 – 0.393

18

18



19

Bean Takeaway

- More Reliant on Pressure Sensors
- Data below Root Zone is Irrelevant
- Larger Variety of Viable Sensors

20

20

Does it work?

- Yes

21

21

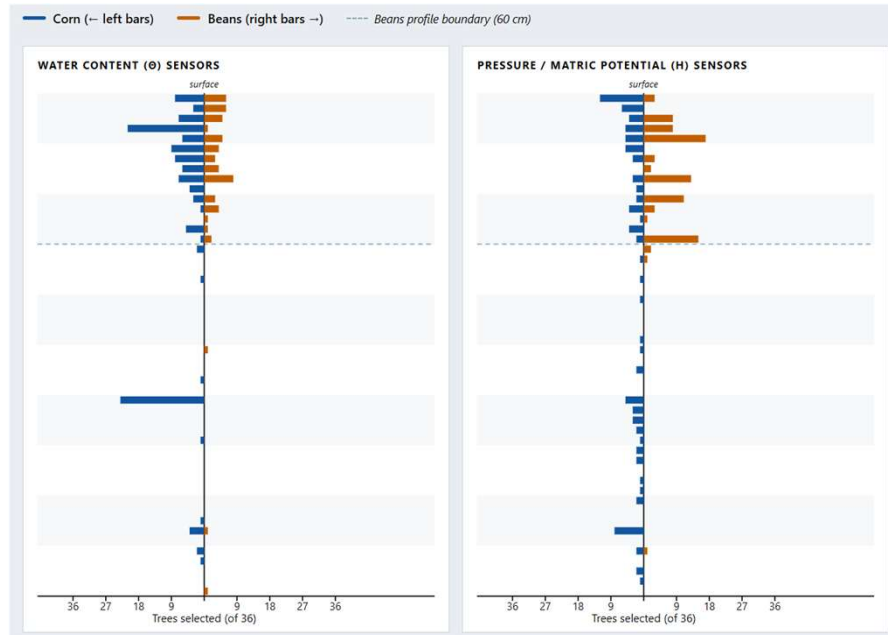
Universal Sensor Layout?

- No

22

22

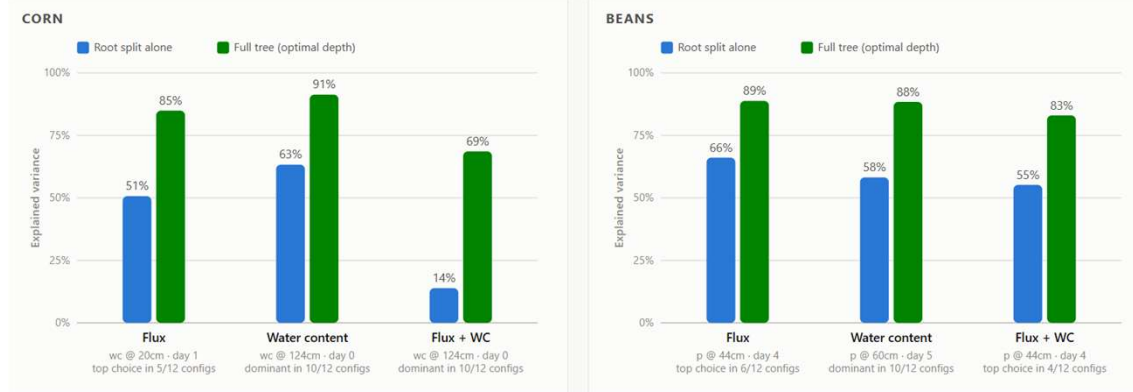
By Crop?



23

23

By Most Useful Sensors?



Crop	Objective	Total Unique Sensors Selected Between All Decision Trees	Selection Rate = 100%	Selection Rate >75%	Selection Rate > 50%	Selection Rate ≤ 50%
Beans	Flux	14	0	0	P @ 20 cm and 36 cm	12
Beans	WC	20	0	P @ 60 cm	p @ 20 cm	18
Beans	Flux+WC	19	0	0	0	19
Corn	Flux	26	0	0	WC @ 16 cm	25
Corn	WC	24	0	WC @ 124 cm	0	23
Corn	Flux+WC	46	0	WC @ 124 cm	WC @ 16 cm	44

24

24

Questions?

References

- Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop evapotranspiration: Guidelines for computing crop water requirements* (FAO Irrigation and Drainage Paper No. 56). Food and Agriculture Organization of the United Nations.
- Ayers, R. S., & Westcot, D. W. (1985). *Water quality for agriculture* (FAO Irrigation and Drainage Paper No. 29, Rev. 1). Food and Agriculture Organization of the United Nations.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. Wadsworth International Group.
- Cameron, K. C., Di, H. J., & Moir, J. L. (2013). Nitrogen losses from the soil/plant system: A review. *Annals of Applied Biology*, 162(2), 145–173. <https://doi.org/10.1111/aab.12014>
- Colmer, T. D., & Voesenek, L. A. C. J. (2009). Flooding tolerance: Suites of plant traits in variable environments. *Functional Plant Biology*, 36(8), 665–681. <https://doi.org/10.1071/FP09144>
- Drew, M. C. (1997). Oxygen deficiency and root metabolism: Injury and acclimation under hypoxia and anoxia. *Annual Review of Plant Physiology and Plant Molecular Biology*, 48, 223–250.
- Evett, S. R., Schwartz, R. C., Casanova, J. J., Heng, L. K., & Baudoin, N. (2012). Soil water sensing for water balance, ET and WUE. *Agricultural Water Management*, 104, 1–9.
- Food and Agriculture Organization of the United Nations. (2014). *The state of food and agriculture 2014: Innovation in family farming*.
- Food and Agriculture Organization of the United Nations. (2023). *The state of food and agriculture 2023*.
- Feddes, R. A., Kowalik, P. J., & Zaradny, H. (1978). *Simulation of field water use and crop yield*. John Wiley & Sons.
- Groenendyk, D. G. (2025). *Texture-naïve soil classification and decision tree-based measurement selection for efficient hydrologic modeling and data collection* (Doctoral dissertation, University of Arizona). <http://hdl.handle.net/10150/678280>
- Guyon, I., & Elisseeff, A. (2003). An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3, 1157–1182.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer.
- Hillel, D. (1996). *Environmental soil physics*. Academic Press.
- Howell, T. A. (2001). Enhancing water use efficiency in irrigated agriculture. *Agronomy Journal*, 93(2), 281–289.
- Israelsen, O. W., & Hansen, V. E. (1962). *Irrigation principles and practices* (3rd ed.). John Wiley & Sons.
- Jones, H. G. (2004). Irrigation scheduling: Advantages and pitfalls of plant-based methods. *Journal of Experimental Botany*, 55(407), 2427–2436.
- Kim, Y., Evans, R. G., & Iversen, W. M. (2008). Remote sensing and control of irrigation. *Computers and Electronics in Agriculture*, 61(1), 44–51.
- Liakos, K. G., Busato, P., Moshou, D., Pearson, S., & Bochtis, D. (2018). Machine learning in agriculture: A review. *Sensors*, 18(8), 2674. <https://doi.org/10.3390/s18082674>
- Lowder, S. K., Sánchez, M. V., & Bertini, R. (2021). Which farms feed the world and has farmland become more concentrated? *World Development*, 142, 105455.
- Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, 12(3), 513–522.
- Murray, D. (2024). *Where the decision tree grows: Exploring the use of tree-based machine learning methods in monitoring well site selection* (Master's thesis, University of Arizona). <http://hdl.handle.net/10150/672786>
- O'Shaughnessy, S. A., Evett, S. R., Colaizzi, P. D., & Howell, T. A. (2019). Using wireless sensor networks for precision irrigation scheduling. *Agricultural Water Management*, 216, 258–268.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weis, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Pereira, L. S., Cordery, I., & Iacovidis, I. (2012). Improved indicators of water use performance and productivity for sustainable water conservation and saving. *Agricultural Water Management*, 108, 39–51.
- Robinson, D. A., Campbell, C. S., Hopmans, J. W., Hornbuckle, B. K., Jones, S. B., Knight, R., Ogden, F., Selker, J., & Wendroth, O. (2008). Soil moisture measurement for ecological and hydrological watershed-scale observatories: A review. *Vadose Zone Journal*, 7(1), 358–389. <https://doi.org/10.2136/vzj2007.0154>

References Cont.

- Schaap, M. G., Leij, F. J., & van Genuchten, M. T. (2001). ROSETTA: A computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. *Journal of Hydrology*, 251(3–4), 163–176.
- Šimůnek, J., van Genuchten, M. T., & Šejna, M. (2008). Development and applications of the HYDRUS and STANMOD software packages and related codes. *Vadose Zone Journal*, 7(2), 587–600.
- Šimůnek, J., van Genuchten, M. T., & Šejna, M. (2016). *Recent developments and applications of the HYDRUS computer software packages*. *Vadose Zone Journal*, 15(7).
- Topp, G. C., Davis, J. L., & Annan, A. P. (1980). Electromagnetic determination of soil water content using TDR: Measurements in coaxial transmission lines. *Water Resources Research*, 16(3), 574–582.
- van Genuchten, M. T. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44(5), 892–898.

27

27

THANK
YOU!

28

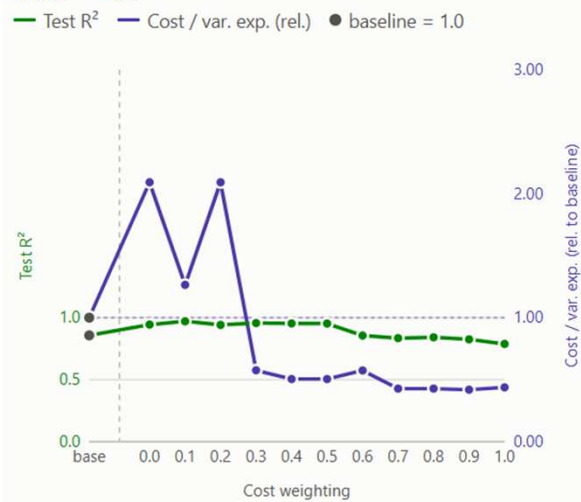
28

Corn — flux

RMSE: 0.0012 – 0.0033 cm/cycle (16 days)
 Observed: -0.0175 – 0.00284 cm/cycle

29

29

Corn — wc

RMSE: 0.0069 – 0.0131
 Observed: 0.198 – 0.341

30

30

Beans — flux

RMSE: 0.007 - 0.028 cm/cycle (8 days)
Observed: -0.219 – 0.057

31

31

Beans — wc

RMSE: 0.0086 – 0.0166
Observed: 0.258 – 0.393

32

32

Measurement Frequency

Timing of Sensor Readings for Corn						
Objective	Number of Trees	Number of Unique Sensors	Early Cycle Readings (day 0-4)	Mid Cycle Readings (day 5-9)	Late Cycle Readings (day 10-15)	Total Readings
Flux	12	78	63	19	53	135
Water Content	12	62	50	16	26	92
Flux+WC	12	186	142	67	81	290
TOTAL	36	244	255	102	160	517

Timing of Sensor Readings for Beans						
Objective	Number of Trees	Number of Unique Sensors	Early Cycle Readings (day 0-2)	Mid Cycle Readings (day 3-5)	Late Cycle Readings (day 6-7)	Total Readings
Flux	12	26	27	30	10	67
Water Content	12	46	52	43	15	110
Flux+WC	12	46	44	27	6	77
TOTAL	36	94	123	100	31	254

33